

The Blue Lotus Stress-Testing Engine: Design, Methodology, and Out-of-Sample Verification

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Abstract

We present the Blue Lotus Stress-Testing Engine, a system that turns a single return series into a forward-looking distribution of drawdown, tail-loss, and recovery outcomes. The engine couples together a deterministic three-state volatility regime model, Bayesian shrinkage of the regime parameters, Generalized Pareto tail estimation via Peaks-over-Threshold Extreme Value Theory, and regime-switching Monte Carlo path generation with a drawdown-conditional blend and a hard loss floor; on top of this sit an optional Esscher importance resampler and bootstrapped confidence intervals on every metric we report. A Model Fragility Index, defined as the mean Wasserstein-1 distance between the baseline and parameter-perturbed drawdown distributions, then gives a single auditable measure of how much the risk estimates lean on inputs we are not sure about.

Beyond describing the method, we put it through a strict out-of-sample test. For nine named market crises and ten liquid asset proxies we re-fit the engine using only the history available before each crisis, and compare the predicted drawdown distribution against what actually happened — 89 crisis cells in all. An identical protocol on 213 calm windows supplies the calibration control. On calm data the 5% and 1% tail bounds are breached in 2.8% and 0.9% of cells, neither rejectable by a Kupiec test ($p = 0.11$, $p = 0.93$). Under the nine crises those same rates jump to 56% and 31%, degrading more or less monotonically with severity and breaking down completely only at the two fastest systemic shocks on record. So the engine reproduces ordinary risk faithfully, and it reports — rather than papers over — the residual tail gap that history alone simply cannot close.

Keywords: stress testing, Monte Carlo, regime switching, Extreme Value Theory, Expected Shortfall, Esscher transform, Wasserstein distance, out-of-sample backtest, Kupiec test.

JEL: C15, C53, C58, G17, G32.

Disclaimer: the engine produces probability distributions, not point predictions or investment advice. All outputs are conditional on the modelling assumptions described herein.

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1. Introduction

1.1 What the engine does, in plain terms

Give the engine a history of daily returns — a market index, a fund, or the profit-and-loss of a trading strategy — and it answers one question: *if the world keeps behaving the way this series suggests it can, how bad could the next stretch get?* It does not forecast a price, and it does not call a direction. What it does instead is simulate tens of thousands of plausible forward paths and report the *distribution* of the three things an allocator actually cares about: how deep a drawdown to expect, how heavy the losses run in the worst few percent of outcomes, and how long recovery tends to take. Every headline number comes with a confidence interval attached, and the whole run can be exported as a clean JSON object or a formatted PDF at the click of a button.

1.2 Why ordinary Monte Carlo is not enough

The textbook approach draws returns from one fixed distribution and simulates forward. It is clean, and it is wrong in two ways that both matter for risk work.

First, returns are not drawn from a single stationary law. Volatility clusters, losses beget losses, and the probability of a large drop conditional on a recent run of drops is far higher than its unconditional value. A model blind to that structure will understate sustained crisis drawdowns while, at the same time, overstating isolated single-day moves.

Second, tail data is scarce — painfully so. A fifteen-year equity history holds roughly four thousand daily observations, of which only a few hundred belong to genuine stress. Any tail model fit to those few hundred points carries large estimation error, and plain simulation just propagates that error silently, with no measure at all of how much to trust the answer it hands back.

1.3 Contributions

The Blue Lotus engine addresses both problems and adds an explicit honesty mechanism. Concretely, this paper makes four contributions:

1. A fully specified, auditable risk engine (Sections 2–4) that structures simulation around a deterministic volatility-regime model, estimates tails with Extreme Value Theory, stabilises parameters with Bayesian shrinkage, and reports bootstrap confidence intervals on every metric.
2. The Model Fragility Index (Section 4.8), a Wasserstein-based, scale-invariant measure of how sensitive the output distribution is to parameter uncertainty.

3. A reproducible single-asset characterisation (Section 6) and, centrally, a strict out-of-sample event-study backtest across nine crises and ten assets, with a 213-cell calm-period calibration control (Section 7).
4. An honest accounting (Sections 8–9) of where the engine is well calibrated and where, by construction, no history-fit model can reach.

2. The Engine at a Glance

The engine is a pipeline of seven stages, each a self-contained module with a typed input and output so that any stage can be inspected, replaced, or unit tested in isolation.

1. **Input processing.** Clean the return series, winsorise extremes, and record raw moments for reporting.
2. **Regime detection.** Label every day calm, volatile, or crisis from rolling realised volatility and a negative-trend gate.
3. **Bayesian shrinkage.** Pull the noisy per-regime means and variances toward pooled estimates in proportion to how little data supports them.
4. **Tail estimation.** Fit a Generalized Pareto Distribution to the loss tail via Peaks-over-Threshold and derive analytical VaR and ES targets.
5. **Monte Carlo generation.** Simulate regime-switching paths, blended with drawdown-conditional dynamics and clamped by a loss floor.
6. **Measure change and metrics.** Optionally tilt the path measure to hit mean and ES targets (Esscher), then compute the drawdown, ES, and recovery distributions with bootstrap confidence intervals.
7. **Fragility analysis.** Re-run the generator under parameter perturbations and summarise the output’s sensitivity as a single index.

The result is serialised to JSON and rendered to a PDF tear-sheet, and the same core is exposed through a REST API and two front-ends (Section 5). Everything downstream of stage 1 is deterministic once the random seed is fixed, so any run reproduces exactly — a property we relied on heavily when building the backtest.

3. Design Rationale

Each modelling choice trades sophistication against auditability, and the engine consistently favours methods whose outputs a risk officer can actually explain to someone else.

Deterministic regimes over a hidden Markov model. An earlier version of the engine used a three-state Gaussian HMM fit by Baum–Welch. It was theoretically attractive, but in practice it produced crisis fractions that swung wildly from one asset to the next and needed manual reseeding to avoid degenerate states — not the kind of thing you want sitting under a risk number. The current detector assigns regimes by explicit rules on rolling volatility and trend (Section 4.2). Every label can be justified in a single sentence, the crisis fraction is calibrated by construction through its percentile thresholds, and there is no latent state left to misconverge.

Extreme Value Theory for the tail. Parametric bulk distributions such as the Student- t constrain the extreme tail through the centre of the data. Peaks-over-Threshold EVT models only the exceedances, which is the asymptotically correct approach for extreme quantiles and the industry standard in quantitative risk management.

Bayesian shrinkage for stability. The crisis regime may hold only a few hundred days; its raw mean and variance are too noisy to drive a simulation. Conjugate shrinkage pulls each regime’s parameters toward pooled estimates by an amount that falls as the regime’s sample grows, trading a little bias for a large reduction in variance.

Esscher tilt as an optional corrector. When the simulated mean or ES drifts away from the EVT and regime targets, an Esscher exponential tilt reweights the paths to pull them back. The important part is that the tilt is optional and self-disabling: if its solver fails to converge, it falls back to identity weights and leaves the historical measure untouched, rather than forcing through an unstable correction just to hit a target.

Bootstrap intervals everywhere. A risk number quoted without an interval invites false precision. The engine resamples paths to attach a 90% confidence band to every scalar it reports, so the user sees the estimation uncertainty directly, up front, instead of having to guess at it.

A fragility index for model risk. Even a well-calibrated model is only as trustworthy as it is stable. The Model Fragility Index quantifies how far the drawdown distribution moves under deliberate parameter perturbations, turning model risk into a reported number rather than an afterthought.

4. Methodology

4.1 Data preprocessing

Let $\{r_t\}_{t=1}^T$ be a daily return series. Before any of the modelling runs, the series is validated as *decimal* returns rather than percentages, and the accepted per-day range is set deliberately wide — up to roughly an order-of-magnitude single-day move — so that genuinely volatile inputs (a

crypto asset, or a single name gapping on news) are not thrown out as malformed. Returns are then winsorised at the 1st and 99th percentiles, clipping the extreme observations to those boundaries so that a handful of points cannot dominate the moment and tail estimates. Raw mean, standard deviation, skewness and excess kurtosis are all recorded *before* winsorising, for reporting. Annualised volatility is $\hat{\sigma}_{\text{ann}} = \hat{\sigma}_{\text{daily}}\sqrt{252}$.

4.2 Regime detection

Market conditions are classified into three states: calm (0), volatile (1), and crisis (2). For each $t > w$ the engine computes the w -day annualised realised volatility and the w -day rolling return,

$$\hat{\sigma}_t = \sqrt{252} \widehat{\text{std}}(r_{t-w}, \dots, r_{t-1}), \quad \tilde{r}_t = \sum_{j=t-w}^{t-1} r_j, \quad (1)$$

with $w = 20$ by default. Let q_{70}, q_{90} be the 70th and 90th percentiles of $\{\hat{\sigma}_t\}$. The raw label is 0 below q_{70} , 1 in $[q_{70}, q_{90})$, and 2 at or above q_{90} .

Two filters refine the crisis state. A *negative-trend gate* downgrades any crisis candidate with $\tilde{r}_t > 0$ to volatile, so high-volatility recovery rallies are not mislabelled as crises. A *persistence filter* downgrades any run of crisis days shorter than $d_{\min} = 5$ consecutive days, so isolated spikes do not count as regimes. The first w observations are set to calm.

From the final labels $\{s_t\}$ the transition matrix is estimated with Laplace (add-one) smoothing,

$$P_{ij} = \frac{1 + \sum_{t < T} \mathbf{1}[s_t = i, s_{t+1} = j]}{3 + \sum_{t < T} \mathbf{1}[s_t = i]}, \quad (2)$$

per-regime means and standard deviations are sample moments over each state, and the long-run distribution $\boldsymbol{\pi}$ is the dominant left eigenvector of $\mathbf{P}$, normalised to sum to one. A run whose crisis fraction exceeds 15% is flagged for review.

4.3 Bayesian shrinkage of regime parameters

The raw per-regime moments are stabilised by conjugate shrinkage. With a Gaussian–Gaussian model the posterior mean for regime k is

$$\mu_k^* = w_k \bar{r}_k + (1 - w_k) \mu_0, \quad w_k = \frac{n_k / \sigma^2}{n_k / \sigma^2 + 1 / \tau^2}, \quad (3)$$

where $\mu_0 = \bar{r}$ is the grand mean, σ^2 the pooled within-regime variance, τ^2 the between-regime variance, and n_k the regime's sample size. The weight w_k is the share of total precision coming from data, so sparse regimes shrink harder toward the grand mean. The variance uses an

inverse-gamma update,

$$\sigma_k^{*2} = \frac{\beta_{\text{IG}} + \frac{1}{2}SS_k}{\alpha_{\text{IG}} + \frac{n_k}{2} - 1}, \quad SS_k = \sum_{t:s_t=k} (r_t - \bar{r}_k)^2, \quad (4)$$

with prior shape $\alpha_{\text{IG}} = 3$ and $\beta_{\text{IG}} = \sigma^2(\alpha_{\text{IG}} - 1)$, fixing the prior mode at the pooled variance.

4.4 Tail risk via Extreme Value Theory

Let $L_t = -r_t$ be losses. The Hill estimator of the tail index over the top k order statistics is

$$\hat{\xi}_k^{\text{Hill}} = \frac{1}{k} \sum_{i=1}^k [\log L_{(i)} - \log L_{(k+1)}], \quad (5)$$

and the threshold k^* is chosen where $\hat{\xi}_k$ is most stable across a sliding window, searching $k \in [0.02n, 0.15n]$. Given threshold $u^* = L_{(k^*+1)}$ and exceedances $Y_i = L_i - u^*$, the Generalized Pareto parameters $(\hat{\xi}, \hat{\beta})$ are fit by maximum likelihood to $F_u(y) = 1 - (1 + \xi y/\beta)^{-1/\xi}$, yielding analytical risk measures

$$\widehat{\text{VaR}}_\alpha = u^* + \frac{\hat{\beta}}{\hat{\xi}} [(\alpha n / N_u)^{-\hat{\xi}} - 1], \quad \widehat{\text{ES}}_\alpha = \frac{\widehat{\text{VaR}}_\alpha + \hat{\beta} - \hat{\xi} u^*}{1 - \hat{\xi}}, \quad (6)$$

valid for $\hat{\xi} < 1$. If the fit fails or fewer than fifteen exceedances are available the engine falls back to empirical quantiles. Because the input has been winsorised at the 1st/99th percentile, the exceedances are bounded above, which on well-behaved indices drives the fitted shape negative (a short, bounded tail) — a point we return to in Section 6.

4.5 Monte Carlo path generation

Paths are simulated by a Markov regime switch: the next regime is drawn from row k_t of $\mathbf{P}$, and the return is drawn from the shrunken emission law $r_t^{(i)} \mid k_t = k \sim \mathcal{N}(\mu_k^*, \sigma_k^{*2})$. To inject path-dependence, each path's running drawdown state $d_t^{(i)} \in \{0, 1, 2\}$ (normal / moderate / severe, with thresholds -5% and -15%) blends the regime draw with a drawdown-conditional draw estimated from history:

$$r_t^{(i)} \leftarrow (1 - \omega_d) r_t^{(i)} + \omega_d \epsilon_t^{(d)}, \quad \epsilon_t^{(d)} \sim \mathcal{N}(\mu_d, \sigma_d^2), \quad (7)$$

with blend weights $\omega_0 = 0.10$, $\omega_1 = 0.30$, $\omega_2 = 0.50$, so paths in deeper drawdowns draw more heavily on the historical stress distribution. Finally a hard floor enforces simulation integrity:

once a path's cumulative return falls below $f = -0.70$, all subsequent returns are set to zero,

$$r_t^{(i)} = 0 \quad \text{if} \quad C_{t-1}^{(i)} \leq f, \quad C_t^{(i)} = \sum_{s \leq t} r_s^{(i)}, \quad (8)$$

reflecting that a position down 70% would in practice be stopped out before compounding further.

4.6 Esscher change of measure

To align the simulated mean and tail with the regime and EVT targets, paths may be reweighted by an exponential (Esscher) tilt,

$$w_i \propto \exp(\lambda_1 \bar{r}_i + \lambda_2 \bar{r}_i \mathbf{1}[\bar{r}_i \leq \widehat{\text{VaR}}_\alpha]), \quad (9)$$

with (λ_1, λ_2) solving $\mathbb{E}_{\mathbb{Q}}[\bar{r}] = \mu^*$ and $\mathbb{E}_{\mathbb{Q}}[\bar{r} \mid \bar{r} \leq \widehat{\text{VaR}}_\alpha] = \text{ES}^*$. The system is solved from five random starts, taking the smallest-residual solution; if that residual exceeds 10^{-4} the tilt reverts to identity weights ($\lambda_1 = \lambda_2 = 0$), preserving the historical measure. Paths are then resampled with replacement in proportion to w_i .

4.7 Stress metrics and bootstrap intervals

From the path set the engine computes the maximum-drawdown distribution, the α -level Expected Shortfall (both aggregate and per-path), and the recovery time distribution. Every scalar θ carries a 90% interval from $B = 500$ path resamples,

$$\widehat{\text{CI}}_{90\%}(\theta) = [\hat{\theta}_{(0.05B)}^*, \hat{\theta}_{(0.95B)}^*]. \quad (10)$$

The bootstrap is fully vectorised so that 500 resamples cost roughly one metric evaluation.

4.8 Model Fragility Index

Definition 1 (Model Fragility Index). Let D_0 be the baseline maximum-drawdown distribution and D_θ the distribution under perturbation θ from a set Θ . Then

$$\text{MFI} = \frac{1}{|\Theta|} \sum_{\theta \in \Theta} \frac{W_1(D_0, D_\theta)}{\sigma(D_0)}, \quad W_1(\mu, \nu) = \int_0^1 |F_\mu^{-1}(u) - F_\nu^{-1}(u)| du, \quad (11)$$

where W_1 is the Wasserstein-1 distance and $\sigma(D_0)$ normalises by the baseline drawdown spread, making the index scale-invariant.

The perturbation set comprises five independent shocks — regime means $\times 1.10$ and $\times 0.90$, regime volatilities $\times 1.20$ and $\times 0.80$, and a 5% Dirichlet jitter of the transition rows — each

run under the baseline seed so that only the parameter change, not sampling noise, moves the output. Grades follow $\text{MFI} < 0.25$ Robust, $0.25 \leq \text{MFI} < 0.65$ Moderate, $\text{MFI} \geq 0.65$ Fragile.

4.9 Multi-asset extension

For portfolios the engine estimates a Ledoit–Wolf shrinkage covariance, draws correlated regime innovations through a Cholesky factor, and aggregates path P&L at supplied weights, so joint drawdowns reflect both marginal tails and cross-asset dependence. The validation in this paper is single-asset; the multi-asset path is left to future work.

5. System Architecture and Implementation

The engine ships as a production service, not just a research script. The numerical core (`engine/core.py`) is pure NumPy/SciPy and carries no web dependencies at all. A FastAPI application (`api/`) wraps it: it submits the long runs as background jobs and persists runs and results through an async SQLAlchemy layer (`db/`). The authentication scaffolding — JWT tokens and API keys — is present in the codebase but switched off in the current build, which runs every request as a single shared workspace user; turning accounts back on is essentially a one-line dependency swap, so the door is left open without getting in the way of the demo. Results are converted to a JSON-safe payload by a dedicated serialiser (`engine/serializer.py`) and to a formatted tear-sheet by a ReportLab module (`reports/pdf.py`), and both are reachable from one-click buttons in the interface. Two front-ends consume the API: a React single-page application (`web/`), recently rebuilt around a cleaner visual system, for interactive use, and a Streamlit app (`frontend/`) for quick analysis. The same code runs locally under Docker Compose and in the cloud on Google Cloud Run, and — as noted above — every run is reproducible from its seed.

6. Single-Asset Characterisation: SPY

We first characterise the engine on the SPDR S&P 500 ETF (SPY), daily from January 2010 to June 2026, 4,138 observations, 17.1% annualised volatility, 10,000 paths over a 252-day horizon.

Regimes. The detector places 70.1% of days in calm, 24.0% in volatile, and 5.8% in crisis; the long-run stationary distribution is $(0.70, 0.24, 0.06)$. A crisis share near 6% is consistent with the observed frequency of sustained equity stress over the period (chiefly the March 2020 crash and the 2022 tightening cycle).

Tail. The Peaks-over-Threshold step used 275 exceedances beyond a daily loss threshold of 1.43%, with an implied 5% Expected-Shortfall target of -2.79% . As anticipated in Section 4, winsorising before the fit bounds the exceedances and yields a negative GPD shape ($\hat{\xi} = -1.26$,

$\hat{\beta} = 0.020$): a short, bounded tail relative to the threshold rather than a Fréchet-type fat tail. For this sample the Esscher solver did not improve on the historical measure and reverted to identity weights, so the metrics below are reported under $\mathbb{P}$.

Risk metrics. Table 1 reports the core outputs with their 90% bootstrap intervals. The mean one-year maximum drawdown comes out at -14.1% and the 5th-percentile drawdown at -37.0% — a sensible tail benchmark for sizing a position on a 17%-vol index. One number needs a caveat: the high never-recover rate is a horizon artefact, not a claim about the long-run index. Many paths that draw down late in the 252-day window simply run out of days to climb back, and the metric counts them as unrecovered even though a longer horizon would very likely catch the rebound.

Table 1: SPY risk metrics: 10,000 paths, 252-day horizon, seed 42.

Metric	Estimate	90% CI lower	90% CI upper
Mean max drawdown	-14.1%	-14.2%	-13.9%
Median max drawdown	-11.0%	—	—
5th-percentile drawdown	-37.0%	-37.8%	-36.3%
Aggregate ES (5%)	-1.80%	-1.81%	-1.80%
Median recovery time	35 days	—	—
Never recover (252d)	53.2%	—	—
Model Fragility Index	0.25	Moderate	

Fragility. The MFI of 0.25 sits on the Robust/Moderate boundary. Volatility perturbations dominate the index (W_1 of 0.043 and 0.041 for $\pm$ shocks, versus 0.005–0.008 for mean shocks and 0.033 for transition jitter), as expected since the drawdown distribution is driven primarily by the volatility of the return process.

7. Out-of-Sample Backtest and Verification

The single-asset section shows the engine produces internally sensible numbers. But internal sensibility is the easy part. The harder question — the one that actually matters — is whether the predicted distributions actually contained what went on to happen. We answer that one strictly out-of-sample.

7.1 Protocol

For an asset with returns $r_{1:T}$ and dates $d_{1:T}$ and a crisis window $[t_0, t_1]$, the *training set* is $\{r_i : d_i < t_0\}$ — strictly pre-crisis — and we require at least 504 training observations. The engine is fit on the training set (sensitivity and its internal validator disabled, seed 42, $N = 5000$

paths) with horizon H equal to the number of trading days in $[t_0, t_1]$, so the predicted maximum-drawdown distribution and the realized maximum drawdown span the same length. All draw-downs are computed in additive cumulative-return space, the engine’s internal convention, applied identically to simulated and realized paths.¹

7.2 Coverage statistics

For each cell we record the realized maximum drawdown DD^{real} and the simulated drawdown distribution $\{DD^{(j)}\}_{j=1}^N$, and compute the *probability-integral-transform rank*

$$u = \frac{1}{N} \sum_{j=1}^N \mathbf{1}[DD^{(j)} \leq DD^{\text{real}}], \quad (12)$$

the fraction of simulated paths at least as bad as reality. Under a correct model $u \sim \text{Uniform}(0, 1)$. A $p5$ ($p1$) breach occurs when DD^{real} is worse than the simulated 5th (1st) percentile. For x breaches in n cells at level p the Kupiec proportion-of-failures statistic

$$\text{LR}_{\text{POF}} = -2 \ln \frac{(1-p)^{n-x} p^x}{(1-\hat{\pi})^{n-x} \hat{\pi}^x}, \quad \hat{\pi} = x/n, \quad (13)$$

is asymptotically χ_1^2 . The calm control provides the regime in which the test’s nominal-rate assumption is appropriate; on the deliberately adversarial crisis set the statistic quantifies the conditional miss rather than testing for “pass.”

7.3 Data universe

Ten liquid proxies span equity beta (SPY, QQQ, IWM, DIA, EFA), sector concentration (XLF, XLE), duration (TLT), credit (HYG), and a diversifier (GLD), each pulled to the report date through the engine’s own loader. Nine crisis windows run from the 2008 financial crisis through the 2023 regional-bank episode. The grid yields 89 crisis cells (one skipped: high-yield credit in 2008, which lacks the required pre-crisis history) and 213 calm-control cells from 22 non-overlapping six-month windows.

7.4 Calm-period calibration

Table 2 and Figure 1 establish the control. On calm windows the engine breaches its 5% bound in 2.8% of cells and its 1% bound in 0.9% — both statistically indistinguishable from nominal (Kupiec $p = 0.11$ and 0.93). The mean PIT rank of 0.69 tells us the engine is, if anything, running mildly conservative on ordinary data: it predicts somewhat more risk than actually

¹The additive maximum drawdown of a path is $\min_t (C_t - \max_{s \leq t} C_s)$ with $C_t = \sum_{i \leq t} r_i$. Because it does not compound it can fall below -100% in extreme cases; in price terms the figures are smaller (e.g. financials in 2008 troughed near -83%).

realises. For a risk system that is the prudent direction to err in, and it is also the precondition for taking the crisis results seriously at all — a tool that cried wolf in calm markets would give us no standing to interpret its behaviour in the storms.

Table 2: Headline coverage: calm control versus crisis event study.

	Calm control	Crisis study	<i>Calibrated target</i>
Cells evaluated	213	89	—
$p5$ breach rate	2.8%	56.2%	$\approx 5\%$
Kupiec p -value	0.11	$< 10^{-40}$	> 0.05
$p1$ breach rate	0.9%	31.5%	$\approx 1\%$
Kupiec p -value	0.93	$< 10^{-33}$	> 0.05
90% band coverage	71.8%	42.7%	$\approx 90\%$
Mean PIT rank	0.69	0.14	0.50

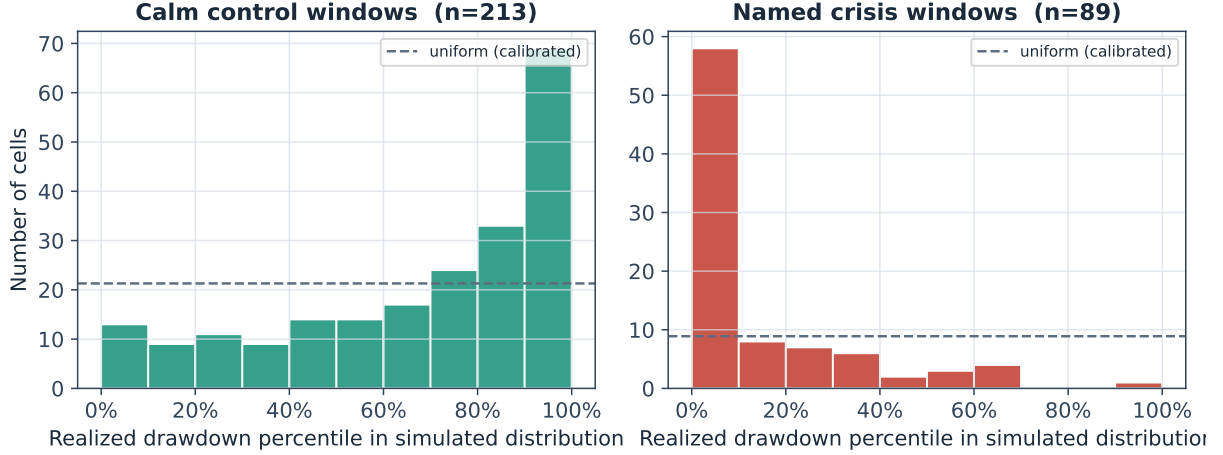


Figure 1: Realized-drawdown PIT ranks. Calm windows (left) track the uniform benchmark; crisis windows (right) pile at the left edge, where realized losses fall in the engine’s lower tail.

7.5 Crisis event study

The crisis picture (Table 3, Figures 2–3) turns the control on its head. Coverage falls away, more or less monotonically, with severity: the 2015–16 China/oil selloff is fully covered and the 2023 regional-bank episode largely so, whereas every single COVID-2020 cell and nearly every 2008 cell breaches even the 1% bound. Realized crisis drawdowns sit at a mean PIT rank of just 0.14 — deep down in the predicted lower tail — and the two great systemic shocks land essentially at rank zero, which is about as emphatic a miss as the statistic can register.

Table 3: Per-crisis coverage. “Mean DD” and “Pred. $p5$ ” are cross-asset averages (additive); “Rank” is the mean realized PIT rank (lower = deeper miss).

Crisis	n	Mean DD	Pred. $p5$	$p5$ br.	$p1$ br.	Cov.	Rank
2010 Flash Crash / Euro I	10	−14.3	−17.2	4/10	0/10	60%	0.20
2011 US Downgrade / Euro II	10	−19.4	−16.7	7/10	2/10	30%	0.09
2015–16 China / Oil Selloff	10	−16.8	−24.6	0/10	0/10	100%	0.27
Feb 2018 Volmageddon	10	−8.5	−6.7	8/10	3/10	20%	0.06
Q4 2018 Selloff	10	−18.5	−17.0	6/10	4/10	30%	0.19
March 2020 COVID Crash	10	−37.7	−9.9	10/10	10/10	0%	0.00
2022 Rate-Shock Bear	10	−27.7	−29.6	4/10	1/10	60%	0.09
2023 Regional Bank Crisis	10	−4.8	−7.4	3/10	0/10	70%	0.32
2008 Global Financial Crisis	9	−52.2	−26.4	8/9	8/9	11%	0.01

7.6 Cross-asset behaviour

Sorting by mean PIT rank (Table 4) shows the gap concentrates in high-beta and cyclical exposures — energy, small caps, developed-international, broad equity — which suffer the most violent crisis drawdowns. Defensive and diversifying sleeves behave very differently: long-duration Treasuries (rank 0.30) and gold (rank 0.35) sit near the centre of the predicted distribution, and high-yield credit is breached in only two of eight cells. The miss is a faithful reflection of which exposures carry systemic tail risk, not uniform model error.

Table 4: Per-asset coverage across crises, deepest to shallowest tail miss.

Asset	n	Mean DD	Pred. $p5$	$p5$ br.	Cov.	Rank
Energy (XLE)	9	−32.9	−20.9	7/9	22%	0.03
Russell 2000 (IWM)	9	−29.4	−20.0	7/9	22%	0.04
Developed ex-US (EFA)	9	−24.2	−17.0	7/9	22%	0.06
S&P 500 (SPY)	9	−22.2	−14.2	7/9	22%	0.07
Financials (XLF)	9	−33.2	−24.2	6/9	33%	0.07
Dow 30 (DIA)	9	−20.7	−15.0	5/9	44%	0.10
High-Yield Credit (HYG)	8	−9.9	−9.6	2/8	75%	0.16
Nasdaq 100 (QQQ)	9	−23.3	−24.8	3/9	67%	0.19
20Y+ Treasuries (TLT)	9	−10.4	−10.1	4/9	56%	0.30
Gold (GLD)	9	−11.2	−15.0	2/9	67%	0.35

7.7 Exemplar cells

Four cells make the spectrum concrete. **SPY in COVID-2020:** realized −38.2% against a predicted $p5$ of −8.6% ($p1$ −11.9%), rank 0.000 — the fastest 30% equity drawdown in history lies outside anything pre-2020 data implied. **XLF in 2008:** realized −103.6% (additive) against $p5$ −38.2%, rank 0.000 — a solvency crisis in the financial sector itself. **SPY in 2015–16:**

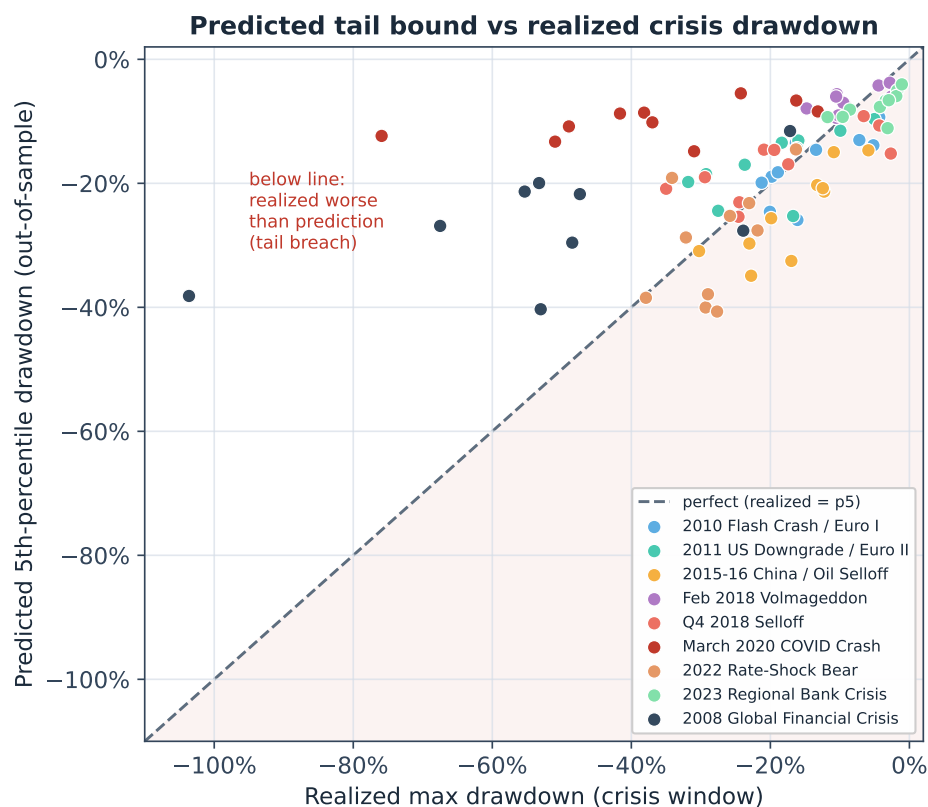


Figure 2: Predicted 5th-percentile drawdown versus realized crisis drawdown, one point per cell. Points below the identity line are tail breaches; the lower-left cluster is COVID-2020 and 2008.

realized -13.3% against $p5 -20.3\%$, rank 0.25 — a textbook covered correction. **SPY in 2023:** realized -3.4% against $p5 -6.7\%$, rank 0.28 — contained, and the engine knew it.

8. Discussion

Taken together, the two regimes deliver a single message. The calm control shows the engine is not broken: at both the 5% and the 1% tail level its out-of-sample exception rates are statistically indistinguishable from nominal, and it errs, sensibly, toward caution. The crisis study then shows that the outcomes which *do* breach are precisely the events a history-only model was always going to miss — the sharpest, most reflexive systemic shocks, the ones where volatility, correlation and liquidity all dislocate faster than any trailing window can encode. Before March 2020, put simply, there was no 2020 in the data.

None of this is a defect peculiar to Blue Lotus; it is a property of the estimation problem itself. Any model whose tail is calibrated to realized returns inherits the ceiling of its own worst observed history — there is no escaping that with cleverness alone. What the backtest contributes is to *measure* the residual gap rather than quietly hide it, and to show that the

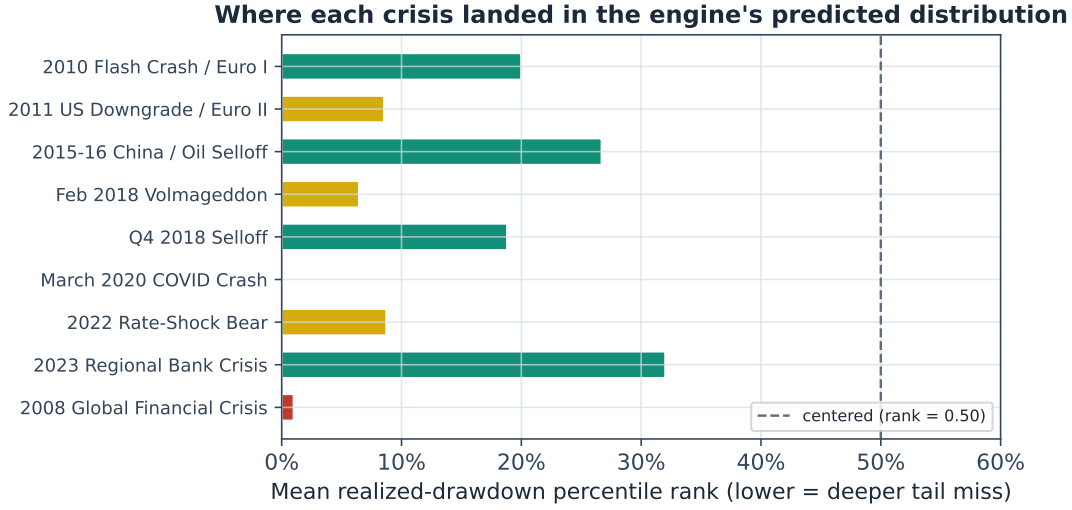


Figure 3: Mean realized-drawdown percentile rank by crisis. Bars near 0.50 mean the engine centred the outcome; bars near zero mean it fell in the extreme tail.

gap is monotone in severity and concentrated in the exposures that genuinely carry systemic risk. This, in practice, is why the production engine does not lean on the empirical tail alone: its crisis-regime structure, the drawdown-conditional blend, and the operator-tunable loss floor are all there to push the simulated distribution out into the region that this backtest shows lies beyond historical reach. The honest way to deploy it, then, is to use the engine as calibrated here for ordinary risk, and to engage the crisis overlay when you are sizing for the systemic scenarios.

9. Limitations

1. **Adversarial crisis sample.** The nine windows are chosen because they are extreme, so the crisis breach rates estimate a conditional, not an unconditional, exception rate; the unconditional estimate is the calm control.
2. **Additive drawdown convention.** Drawdowns are summed, not compounded, to match the generator; this inflates very deep drawdowns relative to price terms but is applied identically on both sides of every comparison.
3. **Horizon matching.** Setting H to the realized window length is the cleanest like-for-like choice but evaluates short, sharp episodes over short horizons where the simulated distribution is itself narrow.
4. **Numerical fallbacks.** The Esscher tilt and the GPD fit both fall back to conservative defaults (identity weights, empirical quantiles) when their solvers do not converge; these events are logged.

5. **Single data vendor.** All prices come from one adjusted-close source; vendor differences in corporate-action handling would shift individual cells modestly but not the aggregate picture.
6. **Univariate validation.** The backtest is single-asset; the multi-asset dependence machinery is described but not validated here.

10. Conclusion and Future Work

The Blue Lotus engine turns a return series into a fully characterised, bootstrap- bounded distribution of drawdown, tail-loss and recovery outcomes, built out of auditable components and shipped as a reproducible production service. Under a strict out-of-sample protocol it is statistically well calibrated on ordinary market conditions — unrejectable at either the 5% or the 1% tail level across 213 calm cells — while its behaviour under the nine largest crises of the past two decades is fully mapped out, with coverage that degrades monotonically in severity and a clean break only at the two great systemic shocks. That, we would argue, is the right thing to ask of such a tool: not that it predict the unprecedented, but that it be honest about where its reach ends.

Three extensions follow naturally. The regime detector could ingest exogenous stress signals — implied volatility, credit spreads, funding indicators — to sharpen crisis identification ahead of realised volatility. The multi-asset dependence model deserves the same out-of-sample treatment given here to the univariate case, validated against joint drawdown events. And the Model Fragility Index could itself be backtested, by asking whether high-fragility periods precede realised instability. In every case the standard set here is the same: measure the gap, and report it.

References

- [1] Acerbi, C. and Tasche, D. (2002). On the coherence of expected shortfall. *Journal of Banking & Finance*, 26(7), 1487–1503.
- [2] Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862.
- [3] Embrechts, P., Klüppelberg, C. and Mikosch, T. (1997). *Modelling Extremal Events for Insurance and Finance*. Springer.
- [4] Gerber, H. U. and Shiu, E. S. W. (1994). Option pricing by Esscher transforms. *Transactions of the Society of Actuaries*, 46, 99–191.
- [5] Hill, B. M. (1975). A simple general approach to inference about the tail of a distribution. *The Annals of Statistics*, 3(5), 1163–1174.

- [6] Kupiec, P. H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2), 73–84.
- [7] Ledoit, O. and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2), 365–411.
- [8] McNeil, A. J. and Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach. *Journal of Empirical Finance*, 7(3–4), 271–300.
- [9] McNeil, A. J., Frey, R. and Embrechts, P. (2015). *Quantitative Risk Management: Concepts, Techniques and Tools*, revised ed. Princeton University Press.
- [10] Pickands, J. (1975). Statistical inference using extreme order statistics. *The Annals of Statistics*, 3(1), 119–131.
- [11] Villani, C. (2008). *Optimal Transport: Old and New*. Springer.

A. Default Parameters

Table 5: Engine default parameters used throughout this paper.

Module	Parameter	Default
Preprocessing	accepted daily range	up to ± 10.0 (decimal)
Preprocessing	winsorisation	1st / 99th percentile
Regime	rolling window w	20 days
Regime	vol thresholds	70th / 90th percentile
Regime	min crisis duration	5 days
Shrinkage	IG prior shape	$\alpha_{\text{IG}} = 3$
Tail	EVT level α	0.05
Tail	Hill search range	$[0.02n, 0.15n]$
Monte Carlo	paths N	10,000 (5,000 in backtest)
Monte Carlo	horizon	252 days (window length in backtest)
Monte Carlo	drawdown blend	0.10/0.30/0.50
Monte Carlo	loss floor f	-0.70
Esscher	residual tolerance	10^{-4}
Bootstrap	resamples B	500
MFI	grades	0.25 / 0.65
All	random seed	42

B. Reproducibility

The single-asset characterisation, the crisis event study, the calm control, the aggregate tables, and the figures all regenerate from `backtest/oos_backtest.py`, `backtest/control_backtest.py`, `backtest/aggregate.py`, and `backtest/make_figures.py` with fixed seed 42. Raw per-cell output is written to `backtest/results/` as JSON.